Ana Silva

GIS AUTOMATION · REGULATORY GIS · SATELLITE MONITORING

From pixels to *the full picture.*

For environmental, infrastructure and nature-risk teams: habitat pre-maps before the field survey, site screening for sensitive locations, and AI-assisted GIS workflows.



ABOUT

Satellite evidence for the people who sign the reports.

I grew up in Porto and now live in Sofia. The path so far runs from cartography and cadastre, through environmental assessment with Natura 2000 habitat mapping and River Habitat Survey reporting, to satellite analytics.

Every sample is built on open data and says what the data can show and what it can't. A satellite map is an input for the ecologist, not a survey result.

SERVICES

Habitat pre-mapping and monitoring

Desk-based habitat maps from Sentinel-1 and Sentinel-2 before the field survey, in UK or EU habitat types, with a parcel table to check on site. Repeat imagery between monitoring surveys.

For: ecological consultancies, habitat bank operators

Site screening for sensitive locations

Sites screened against protected areas, critical habitat, water risk and land cover, on open data, shaped for IFC PS6, EBRD PR6 and the TNFD Locate phase.

For: ESIA consultancies, ESG and nature-risk teams

AI-assisted GIS workflows

Guidance and training on what to hand to AI assistants in geospatial work and what to check: projections, areas, units, validation.

For: GIS teams starting with AI tools

PATH

- **Cartography & cadastre**
  where every boundary had to be right
- **Environmental assessment**
  Natura 2000 habitat mapping, River Habitat Survey reporting
- **Satellite analytics**
  large datasets, time series, machine learning
- **Freelance**
  support for ecologists and consultants, AI-assisted GIS training

WORKED WITH

Multinational food, beverage and retail companies, agricultural and forestry producers in Brazil, agricultural insurers, and the World Health Organization.

TRAINING

FPIF certified trainer (2018) · classes at Universidade Júnior

TOOLS

Python (xarray, dask, geopandas, scikit-learn) · QGIS · Google Earth Engine · Claude Code · Antigravity · n8n

LANGUAGES

Portuguese · English · Bulgarian · Spanish (comprehension)

BASED IN

Sofia, Bulgaria

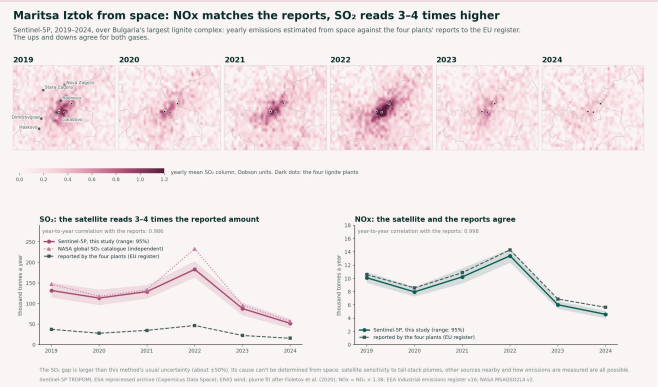
Have land you need to understand?

Freelance work is contracted through Timebooster (EU VAT registered).

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https://kirjawa.github.io

Selected work

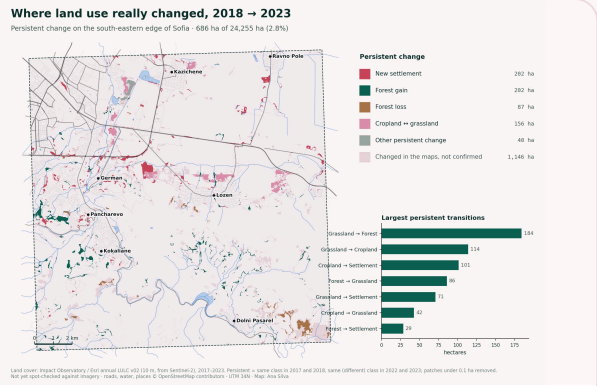
Demonstrations on open data, never client work



COMPLETE · MARITSA IZTOK, BULGARIA

Maritsa Iztok: do the emissions seen from space match the reports?

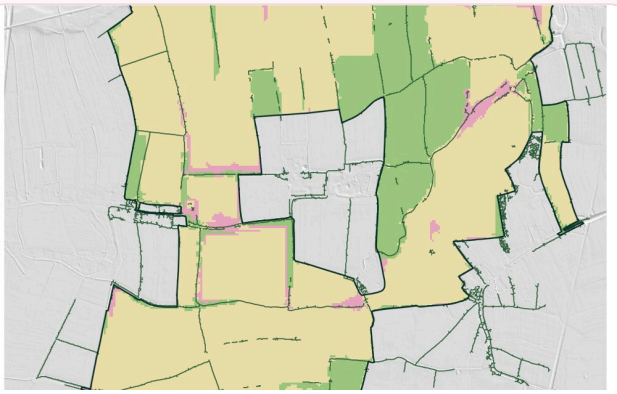
Sentinel-5P over Bulgaria's largest lignite complex, 2019 to 2024. The ups and downs match the plants' reports for both gases. NOx matches the reported amount too; SO_2 reads 3 to 4 times higher, as NASA's independent estimate also finds.



COMPLETE · SOFIA, BULGARIA

Land-use change that lasts

686 ha of persistent change 2018–2023 on Sofia's south-eastern edge, including 202 ha of new settlement. Year-to-year map noise filtered out.



SAMPLE · UNITED KINGDOM

Habitat pre-map and monitoring for biodiversity net gain

A desk-based pre-map of a consented solar farm site from open satellite data and LiDAR, checked against its published survey: 75% cropland and 20% grassland against 71% and 26%, the site's record since 2017, and a table for the statutory metric.



SAMPLE · BULGARIA AND PORTUGAL

Site screening for lender standards and nature disclosures

1,237 solar and wind sites screened on open data: 276 would likely need a critical habitat assessment.



AGRICULTURE · WITH ATLASTICO

Crops, fields and supply chains

Crop monitoring, phenology, soil cover, drought and palm oil sourcing, including Season's Pulse. Taken on through Atlastico: kirjawa.github.io/agriculture.html